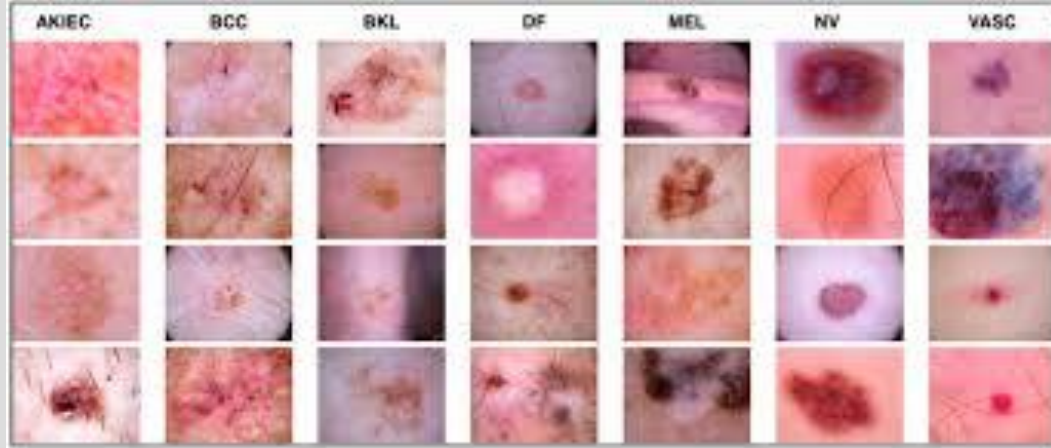


Automatic Image Analysis

Topic B: Bayesian Decision and Uncertainty: Skin Lesion Triage

Presentation 2: Modern pipeline & Comparison



Group 20:

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Furkan Akgün

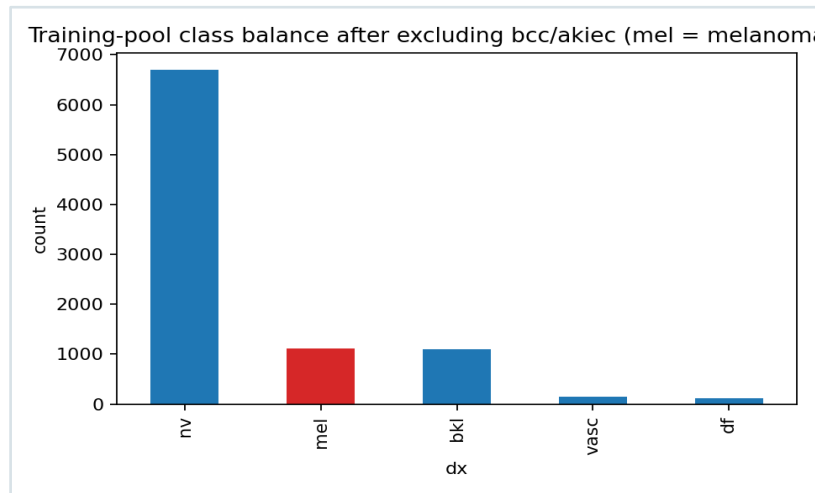
RECAP

Classical Baseline & the Core Question

Classical Approach

- Convert dermoscopy images to HSV, extract per-channel histograms (32 bins)
- Fit a Gaussian Mixture Model per class (K via BIC: 28 benign, 8 melanoma)
- Apply Bayes' rule for posteriors, then a MAP vs. cost-sensitive decision

Main question: how should the classifier behave when missing a melanoma is far more costly than a false alarm?



Training-pool class balance (melanoma in red)

1,860

held-out test lesions

10.8%

melanoma prevalence

0.143

classical ECE

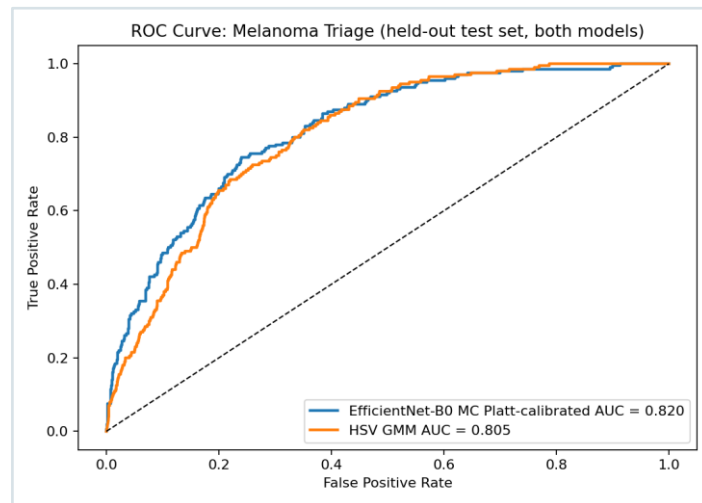
54%

melanomas <1% risk

EfficientNet-B0 with a Fine-Tuned Head

Architecture & Training

- ImageNet-pretrained EfficientNet-B0 backbone + new classification head
- Stage 1: freeze backbone, train head only (6 epochs)
- Stage 2: unfreeze last 2 blocks, fine-tune at low LR (6 epochs)
- ~7 minutes total training on a single Kaggle T4 GPU



ROC curve, held-out test set ($n = 1,860$)

0.820

EfficientNet test AUC

0.805

classical GMM AUC

+0.015

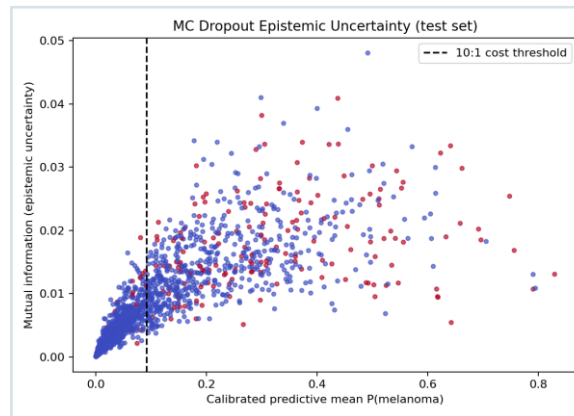
AUC improvement

UNCERTAINTY

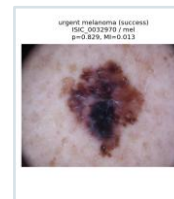
Knowing What the Model Doesn't Know

How It Works

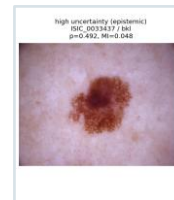
- Keep dropout active at inference; run 30 stochastic forward passes
- Averaging the passes approximates a Bayesian posterior over weights
- Predictive entropy splits into aleatoric (data noise) + epistemic (model unsure)
- High epistemic uncertainty -> route to manual review, not an automatic call



Mean vs. epistemic uncertainty



Confident (low MI)



Uncertain (high MI)

21.2%
auto-reassured

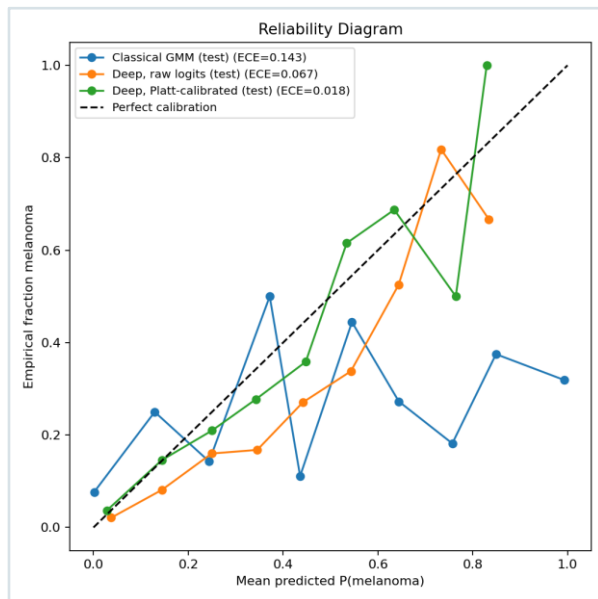
76.2%
manual review

2.6%
urgent referral

98.5%
sensitivity kept

COMPARISON

Whose Confidence Can You Trust?



Closer to the diagonal = better calibrated

Calibration error (ECE, lower is better)

0.143

Classical GMM

0.067

Deep, raw logits

0.018

Deep, calibrated

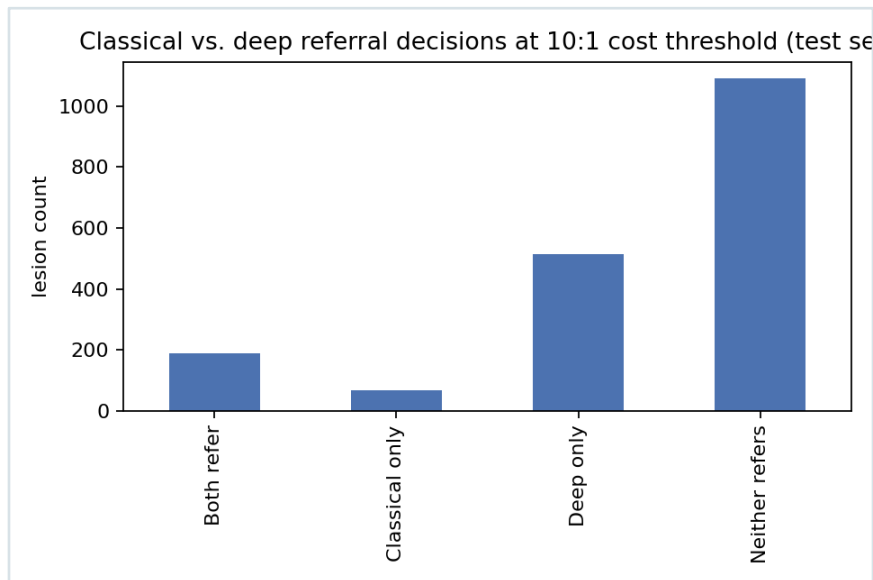
Reaching safety-critical sensitivity (95%+)

Model	Sensitivity	Referral needed
Classical GMM	95%+	100% (all lesions)
EfficientNet-B0	97.5%	68%

Classical GMM's posteriors don't separate classes well enough for safety-critical use; EfficientNet does.

COMPARISON

Classical vs. Deep: Which Model Wins?



Referral decisions at the 10:1 cost threshold

68.8%

agreement
between models

26/200

melanomas missed
by both

514

lesions deep
model alone flags

Recommended: EfficientNet-B0 + cost-sensitive threshold ($\tau = 0.091$) + uncertainty-aware triage.